Title: Explainable Transformer-Based Emotion-Aware Customer Feedback Analysis

Abstract

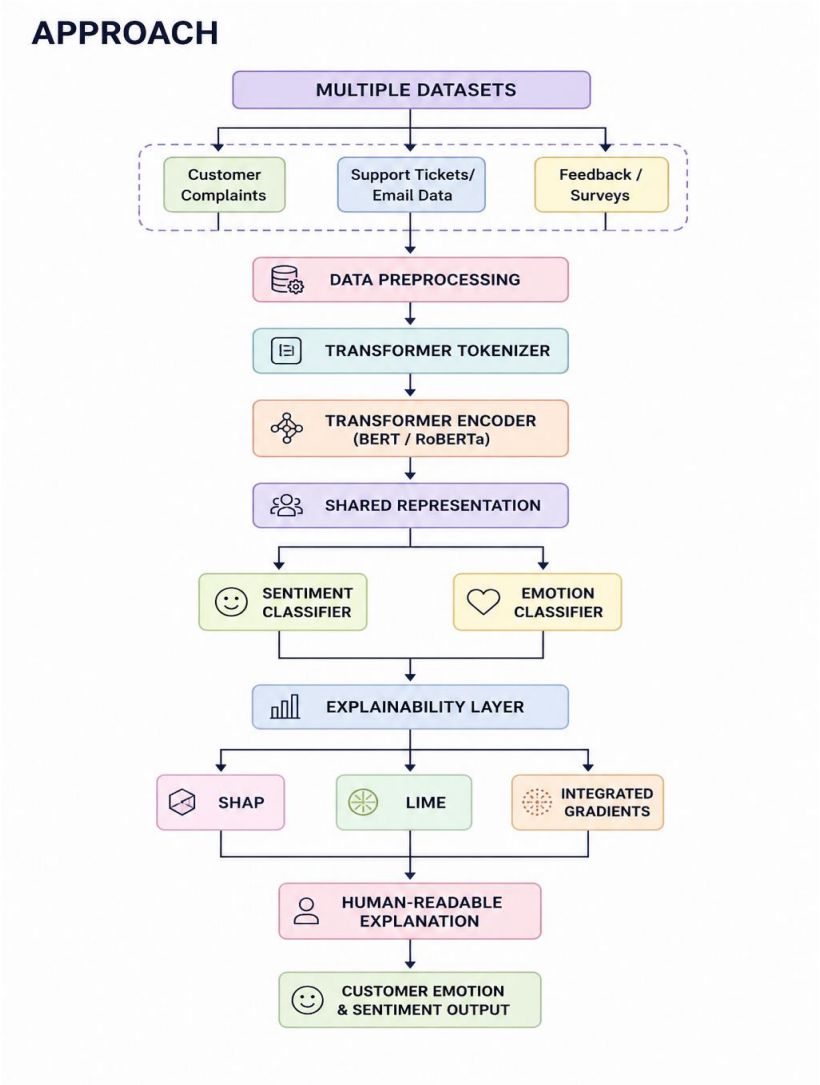
Customer service organizations receive a large number of complaints every day through different communication channels. These complaints are usually provided as unstructured text and need to be categorized into specific classes before they can be processed efficiently. Manual classification of complaints requires significant time and human effort and may also lead to inconsistent categorization. Traditional machine learning techniques such as Naïve Bayes, Logistic Regression, and Support Vector Machines commonly use Bag-of-Words or TF-IDF features, which have limited capability to understand the context and semantic relationships within complaint text.

This project presents a **BERT-based Customer Complaint Classification System** that uses a pre-trained Bidirectional Encoder Representations from Transformers (BERT) model for automated text classification. The model is fine-tuned on a customer complaint dataset so that it can learn domain-specific language patterns and identify the contextual meaning of different complaints. The system takes a customer's complaint as input and predicts its corresponding predefined category. The performance of the proposed model is evaluated using **Accuracy, Precision, Recall, and F1-Score**.

Datasets

By leveraging BERT's bidirectional contextual understanding, the proposed system aims to provide more accurate and reliable complaint classification than conventional text classification approaches. The system can reduce the effort involved in manual categorization and provide an efficient approach for handling large volumes of customer complaints. The project demonstrates the applicability of transformer-based NLP techniques to real-world customer service text classification problems.

Dataset	URL link
Consumer finance Dataset	https://www.consumerfinance.gov/data-research/consumer-complaints
Kaggle dataset	https://www.kaggle.com/datasets/namigabbasov/consumer-complaint-dataset/data
Bitext Customer Support Dataset	https://huggingface.co/datasets/PolyAI/banking77



Project Group Members

S.No.	Roll.No.	Student Name
1	2420080045	SAI JASHAN
2	2420080044	Adithya
3	2420080008	Rohit